The irrationality exponent of $\log 2 + \sqrt[3]{2}$ is at most six

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Abstract

We prove that $\log 2 + \sqrt[3]{2}$ has irrationality exponent at most six. We adapt our previous affine determinant construction to a cubic field with one compatible embedding. Clearing a single minor gives an algebraic integer whose norm compares analytic decay at that embedding with growth at the other two. The norm contains three copies of the clearing factor and two coarse determinant bounds. We choose parameters that absorb this full cost, using the logarithmic interpolation and analytic estimates from our previous proof. A Lean formalization accompanies the argument.

1 Introduction and main result

Let r be the positive real root of $r^3 = 2$ and put $x = \log 2 + r$. The number x is transcendental: the Hermite–Lindemann theorem [5] gives the transcendence of $\log 2$, and adding the algebraic number r preserves it. We study how well x can be approximated by rational numbers.

For an irrational real number y, write $\mu(y)$ for the supremum of the positive exponents ν such that $0 < |y - p/q| < q^{-\nu}$ for infinitely many reduced rational numbers p/q with $q > 0$.

Theorem 1.1. *For every real $\nu > 6$ there exists $Q \in \mathbb{N}$, $Q \geq 2$, such that*

$$q^{-\nu} \leq \left| \log 2 + \sqrt[3]{2} - \frac{p}{q} \right| \qquad (p \in \mathbb{Z},\ q \in \mathbb{N},\ q \geq Q). \tag{1}$$

Consequently, $\mu(x) \leq 6$.

The threshold may depend on ν and is uniform in p, q; the fractions in (1) need not be reduced. The proof gives neither a numerical value of Q nor an estimate at $\nu = 6$.

Our previous affine theorem [1, Theorem 3.1] treats integral data in a degree-four number field with two compatible embeddings. Here we use the same determinant construction in degree three, with one compatible embedding and slope of absolute value at most one. We retain the integrality assumptions. The logarithmic interpolation and analytic estimates come from our previous proof, which builds on OpenAI's determinant method [2] and our logarithmic extensions [3].

The change is in the norm comparison. All three embeddings contribute to the norm, but only one supplies the analytic decay we need. We bound the other two coarsely and choose parameters that absorb both bounds. This works only if we keep the same minor at every embedding: the norm must belong to one algebraic integer. Section 3 sets out the three estimates, Lemma 4.1 supplies that common algebraic integer, and Section 5 chooses the parameters that make the estimates incompatible.

Rational approximation to x can also be viewed as approximation to $\log 2$ by the cubic numbers $p/q - r$. This explains why a bound for rational approximation to $\log 2$ alone does not give Theorem 1.1. Section 6 relates the two height conventions.

2 A cubic affine theorem

For a number field F, write $\mathrm{Emb}(F) = \mathrm{Hom}_{\mathbb{Q}\text{-alg}}(F, \mathbb{C})$ and let $\mathcal{O}_F$ be its ring of integers.

Theorem 2.1. *Suppose $[F : \mathbb{Q}] = 3$ and $a, u, v \in \mathcal{O}_F$. Let $\sigma_0 \in \mathrm{Emb}(F)$ have $\sigma_0(a) = b_0 \in \mathbb{R}$, where $b_0 > 0$ and $b_0 \neq 1$. Let $y \in \mathbb{R} \setminus \{0\}$, and suppose an embedding $\tau \in \mathrm{Emb}(F)$ satisfies*

$$\exp\bigl(\tau(u)(y - \tau(v))\bigr) = \tau(a), \qquad |\tau(u)| \leq 1. \tag{2}$$

Then for every real $\nu > 6$ there is $Q \in \mathbb{N}$, $Q \geq 2$, such that $q^{-\nu} \leq |y - p/q|$ for all integers p and natural $q \geq Q$.

The selected embedding τ need not equal σ_0. The other two embeddings are bounded coarsely, whether or not either happens to satisfy the same exponential identity. The roles are different: σ_0 gives a nonzero minor by interpolation, while τ gives the analytic upper bound on its determinant.

2.1 Specialization to the target

Take $F = \mathbb{Q}(r)$, $a = 2$, $u = 1$, $v = r$, and $y = x = \log 2 + r > 0$. Eisenstein's criterion at 2 proves that $T^3 - 2$ is irreducible, so $[F : \mathbb{Q}] = 3$. The element r is integral, as are a, u. The real embedding is a permissible choice for both σ_0 and τ, since $\exp(x - r) = 2$.

If ζ is a primitive cube root of unity, the other embeddings send r to $r\zeta$ and $r\zeta^2$. They obey

$$|\exp(x - r\zeta)| = |\exp(x - r\zeta^2)| = 2\exp(3r/2) > 2.$$

Thus they do not supply the identity used at the real embedding, even though every embedding sends the base a to 2. Theorem 2.1 now gives (1); Section 5.1 deduces the exponent bound.

3 The inherited matrix estimates

We first record the estimates we will use from our previous proof [1, Sections 4–8]. The parameters remain fixed as the matrix height H tends to infinity. We choose them in Section 5, after seeing the full cost of the cubic norm.

3.1 Parameters and the selected minor

Fix $\nu > 6$. Assume, for contradiction, that there are approximations

$$|y - p/q| \leq q^{-\nu} \tag{3}$$

with arbitrarily large denominators. We select a finite tuple p_i/q_i with $q_i \geq 2$ and $p_i \neq 0$, and set $w_i = \lceil \log q_i \rceil$, $w_* = \min_i w_i$. The nonzero-numerator requirement causes no loss because $y \neq 0$. Each weight is chosen large enough, after the preceding weights, to satisfy the separation conditions for logarithmic interpolation [1, Sections 4–5]. Arbitrarily large denominators let us make these successive choices.

We take $0 < \eta < 1$ and rational shape parameters satisfying

$$0 < \theta < A < B < 1, \quad C > 1, \quad CB < 1, \quad B < C\theta < 1, \tag{4}$$

For a positive scale $F_0 > 2/\theta$ and dimension $m \geq 1$, set

$$K = \lfloor C^m \rfloor, \qquad w_0 = B^{-m}, \qquad v_0 = 2K\theta^m w_0. \tag{5}$$

The strict volume ratio is $Kw_0\theta^m/v_0 = 1/2$.

The main quantities in the argument have the following roles.

Quantity	Role
$m,\ H$	Number of approximations and matrix height
$K,\ M(H)$	Number of multiplicative centers and number of rows
$w_i,\ w_0,\ v_0$	Approximation weights and the weights of h and s
b_H	Normalized mean row weight saved by denominator clearing
$\mathscr{C}_H,\ \mathcal{C}_H$	Rational clearing factor and analytic collision parameter
$E,\ U,\ V$	Arithmetic error, coarse growth, and analytic error

For $H > 0$, columns are $(h, \gamma) \in \mathbb{N} \times \mathbb{N}^m$ satisfying $w_0 h + \sum_i w_i\gamma_i \leq H$. Rows are (j, s, β) with $0 \leq j < K$ and $v_0 s + \sum_i w_i\beta_i/\theta < H$. Write $M = M(H) > 0$ for the row count. Put

$$T_i = \lceil F_0 w_i/v_0 \rceil, \qquad L_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1}}{k} t^k, \qquad c_{\sigma i} = \sigma(u)(p_i/q_i - \sigma(v)).$$

The matrix at an embedding σ is

$$\mathcal{M}_\sigma(H)_{(j,s,\beta),(h,\gamma)} = \sigma(a)^{jh}[t^s Z^\beta](1+t)^h \prod_{i=1}^{m} (Z_i + jc_{\sigma i} + L_i(t))^{\gamma_i}. \tag{6}$$

At σ_0, the multiplicative centers b_0^j are distinct and nonzero. The interpolation theorem allows arbitrary additive centers, so it gives full row rank at a cofinal set of heights. At each such height we select one nonzero full-row minor and use exactly those columns at every embedding. Write Δ_σ for the resulting determinant. Define the mean saved row weight by

$$b_H = \frac{1}{MH} \sum_{\text{rows } (j,s,\beta)} \sum_i w_i\beta_i, \qquad 0 \leq b_H \leq \theta.$$

3.2 Clearing, growth, and decay

Choose fixed bounds $\kappa, R \geq 0$ and $\rho \geq 100$ such that

$$\begin{aligned} \log \max(1, |\sigma(a)|) &\leq \kappa, \\ |\sigma(u)|(|y| + 1 + |\sigma(v)|) &\leq R \qquad\qquad (\sigma \in \mathrm{Emb}(F)), \\ 100|y| \leq \rho, \qquad 100|\tau(u)(y - \tau(v))| &\leq \rho. \end{aligned}$$

There are finitely many embeddings, so we can fix these bounds before choosing the dimension or approximations. For $x = \log 2 + r$, we may take $\kappa = \log 2$, $R = 6$, $\rho = 300$. Put $\Lambda = \log 4 + 4$ and

$c = \log 2/4$, as in our previous estimates, and define

$$E = \frac{\Lambda F_0 m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta}{w_*}, \tag{7}$$

$$U = \frac{\log 2}{v_0} + \frac{\kappa K + \log(3/2)}{w_0} + \frac{\log(2(KR+2))}{w_*},$$

$$V = \frac{\nu}{F_0} + \frac{2\log 2}{v_0} + \frac{\rho K}{w_0} + \frac{\log 4 + \log(2K) + \nu + \log(2\rho K)}{w_*}.$$

The first estimate controls denominator clearing. It supplies a positive rational factor $\mathscr{C}_H$ with

$$\frac{\log \mathscr{C}_H}{MH} \le 1 - b_H + E. \tag{8}$$

The second bounds growth at each of the other two embeddings: for $\sigma \neq \tau$,

$$\frac{\log |\Delta_\sigma|}{MH} \le U + r_c(H), \qquad r_c(H) \longrightarrow 0. \tag{9}$$

At the selected embedding write $\omega = \tau(u)(y - \tau(v))$. The error comparison is

$$|c_{\tau i} - \omega| = |\tau(u)|\,|p_i/q_i - y| \le q_i^{-\nu}.$$

The slope bound therefore lets us use the third estimate without enlarging the approximation error:

$$\frac{\log |\Delta_\tau|}{MH} \le V + r_a(H) + \max\{-\mathcal{C}_H, -\nu(A(1-\eta) - b_H)\}, \qquad r_a(H) \longrightarrow 0, \tag{10}$$

where

$$\mathcal{C}_H \longrightarrow \mathcal{C}_\infty = \frac{c\eta^2 K \theta^m}{(m+1)v_0 A^m} = \frac{c\eta^2}{2(m+1)}(B/A)^m. \tag{11}$$

These are the estimates of our previous affine construction [1, Sections 6–8], applied to the same selected minor. Notice that the analytic frequency ω and the number y being approximated need not coincide. In our concrete example they are $\log 2$ and $\log 2 + r$, respectively. The identity at τ is enough; we do not take logarithms at the other embeddings.

4 The norm of the common minor

The affine numerators $n_i = u(p_i - q_i v)$ are integral. Set $D_i = \operatorname{lcm}(1, \dots, T_i)$ and $e_i = \lfloor H/w_i \rfloor$. Multiply each row by $\prod_i D_i^{e_i} q_i^{-\beta_i}$ and each selected column by $\prod_i q_i^{\gamma_i}$. Their product is $\mathscr{C}_H$.

Lemma 4.1. *There is one nonzero $Z_H \in \mathcal{O}_F$ such that*

$$\sigma(Z_H) = \mathscr{C}_H \Delta_\sigma \qquad (\sigma \in \operatorname{Emb}(F)).$$

Every selected determinant is consequently nonzero.

Proof. If some $\beta_i > \gamma_i$, the matrix entry is zero. Otherwise put $\delta_i = \gamma_i - \beta_i$. After coefficient extraction and clearing, the entry is the image of

$$a^{jh} \prod_i \binom{\gamma_i}{\beta_i} [t^s](1+t)^h \prod_i D_i^{e_i} (j n_i + q_i L_i(t))^{\delta_i}.$$

The polynomial $D_i(j n_i + q_i L_i)$ has integral coefficients, and $\delta_i \le \gamma_i \le e_i$. Hence this entry is integral. These entries define one matrix over F; let Z_H be its determinant. Determinants commute with every embedding, proving the identity. Its image at σ_0 is nonzero, and every embedding is injective. □

We can now use the norm of Z_H, which is a nonzero integer. Counting all three embeddings into $\mathbb{C}$, with each member of any nonreal conjugate pair counted separately, we obtain

$$0 \le 3\log \mathscr{C}_H + \sum_{\sigma \in \mathrm{Emb}(F)} \log|\Delta_\sigma|.$$

Using (8) and the two bounds (9) gives

$$\frac{\log|\Delta_\tau|}{MH} \ge -3(1-b_H) - 3E - 2U - 2r_c(H). \tag{12}$$

This is where degree three enters the proof. The clearing factor appears three times in the norm, and each of the two other embeddings contributes one coarse bound. A conjugate pair may have equal absolute values, but both factors still occur in the product.

5 Stronger parameters and the contradiction

For every $\nu > 6$, the shape can be chosen so that

$$g = \nu(A(1-\eta) - \theta) - 3(1-\theta) > 0. \tag{13}$$

For example, take rational $t < 1$ close enough to one that $\nu t > 3(1+t)$, put $\theta = t^2$, choose rational $\theta < A < B < t$ sufficiently close to t, then choose $B/\theta < C < 1/B$ and sufficiently small $\eta > 0$. These choices satisfy (4) and (13).

Let $S = 3E+2U+V$. We need $S < g$ to handle the approximation term in (10), and $3+S < \mathcal{C}_\infty$ to handle the collision term. Expanding (7) gives

$$\begin{aligned} S = {} & \frac{\nu}{F_0} + \frac{3\Lambda F_0 m + 4\log 2}{v_0} + \frac{(\rho+2\kappa)K + 2\log(3/2)}{w_0} \\ & + 3\Lambda \sum_i \frac{1}{w_i} + \frac{3\theta + \log 4 + \log(2K) + \nu + \log(2\rho K) + 2\log(2(KR+2))}{w_*}. \end{aligned} \tag{14}$$

Choose $0 < \epsilon < g$ with $\epsilon \le 1/2$, and next choose $F_0 > 2/\theta$ with $\nu/F_0 < \epsilon/3$. As $m \to \infty$ in (5),

$$m/v_0 \to 0, \qquad 1/v_0 \to 0, \qquad K/w_0 \to 0, \qquad 1/w_0 \to 0, \qquad \mathcal{C}_\infty \to \infty.$$

The limits involving v_0 use $C\theta/B > 1$, the ratio K/w_0 tends to zero because $CB < 1$, and the collision parameter grows because $B/A > 1$. Choose one dimension making the two dimension terms of (14) less than $\epsilon/3$ and $\mathcal{C}_\infty > 4$. Choose a positive rational geometric slack satisfying the interpolation conditions in [1, Section 4]. The strict volume ratio $1/2$, $K\theta^m < 1$, and $\theta < 1$ permit this choice. Then choose a sufficiently large initial weight threshold so that the remaining inverse-weight terms sum to less than $\epsilon/3$. Subsequent approximations can still be selected with the separated-weight inequalities of the same section because their denominators are unbounded under (3). All these choices precede the limit in H.

We have obtained

$$S < \epsilon < g, \qquad 3 + S < 4 < \mathcal{C}_\infty. \tag{15}$$

Set $S_H = S + r_a(H) + 2r_c(H)$. For all sufficiently large H, (15) gives $S_H < g$ and $3 + S_H < \mathcal{C}_H$. Combining (10) and (12) yields

$$-3(1-b_H) \le S_H + \max\{-\mathcal{C}_H, -\nu(A(1-\eta) - b_H)\}.$$

If the maximum is the approximation term, then

$$g + (\nu - 3)(\theta - b_H) = \nu(A(1-\eta) - b_H) - 3(1 - b_H) \le S_H < g,$$

a contradiction. If it is the collision term, then $\mathcal{C}_H \le 3(1 - b_H) + S_H \le 3 + S_H$, also a contradiction. Cofinality of full-row-rank heights permits the same selected-minor construction beyond this common threshold. This proves Theorem 2.1.

5.1 Endpoint consequences

The contradiction excludes (3) at all sufficiently large denominators, giving (1). The lower bound also supplies irrationality directly: a rational x would have exact representations with arbitrarily large unreduced denominators. For any exponent above six, reduced fractions satisfying its strict upper bound have bounded denominators, and their numerators are bounded as well. Thus there are only finitely many such fractions, which gives $\mu(x) \le 6$.

5.2 Why six appears

The conditions $B < C\theta$ and $CB < 1$ imply $B^2 < \theta$, hence $A < \sqrt{\theta}$. Consequently

$$\frac{A(1-\eta) - \theta}{1 - \theta} < \frac{\sqrt{\theta}}{1 + \sqrt{\theta}} < \frac{1}{2}.$$

The sufficient gap (13) therefore requires $\nu > 6$. Six is the infimum cutoff for this choice of shape using only $0 \le b_H \le \theta$. To lower it, we would need a different shape or a stronger estimate for the saved row weight. The argument does not determine the exact value of $\mu(x)$.

6 Approximation by cubic numbers

Marcovecchio's rational approximation and nonquadraticity estimates for $\log 2$ [4] concern different families of approximants. In our problem the algebraic shift gives a particular family of cubic numbers. For a reduced fraction p/q, our approximation error is

$$|x - p/q| = |\log 2 - (p/q - r)|.$$

The cubic number $p/q - r$ has primitive irreducible integer polynomial

$$(qT - p)^3 + 2q^3 = q^3T^3 - 3pq^2T^2 + 3p^2qT - p^3 + 2q^3.$$

Irreducibility follows from the rational affine change of variable from $T^3 - 2$. For primitivity, any prime dividing q fails to divide the constant coefficient. Near the fixed target, the coefficient height is bounded above and below by positive fixed multiples of q^3. Thus a general degree-three approximation measure expressed in that height would require a factor-three conversion of its exponent to this rational-denominator convention.

Thus the relevant comparison is with degree-three approximation to $\log 2$ along the family $p/q - r$, after converting its coefficient height to the denominator q. The rational and quadratic estimates in [4] do not give this comparison directly.

A Formalization

The accompanying Lean formalization, `AffineCubic`, proves Theorems 1.1 and 2.1. It constructs the real field $\mathbb{Q}(2^{1/3})$ and proves the irreducibility, degree, integrality, and distinguished-embedding identity needed to apply Theorem 2.1.

The concrete results are in the `Pilot` module. Alongside the eventual lower bound and exponent inequality, it proves irrationality and $\neg\texttt{LiouvilleWith}(\nu, x)$ for every $\nu > 6$, using Mathlib's predicate. The last consequence follows by applying the lower bound at $(6+\nu)/2$, which absorbs the fixed constant allowed in that predicate. The companion `theorem-map.json` identifies the declarations, and `VERIFY.md` gives the source versions, reproduction instructions, and proof-checking results.

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